# Inner Product Spaces
(and other associated crap)

An inner product is a function which associates a real number $\langle \underline{u}, \underline{v} \rangle$ to each pair of vectors $\underline{u}, \underline{v}$ within a vector space, satisfying the following axioms:

$\langle \underline{u}, \underline{v} \rangle = \langle \underline{v}, \underline{u} \rangle$ — Symmetry

$\langle \underline{u}+\underline{v}, \underline{w} \rangle = \langle \underline{u}, \underline{w} \rangle + \langle \underline{v}, \underline{w} \rangle$ — Additivity

$\langle k\underline{u}, \underline{v} \rangle = k \times \langle \underline{u}, \underline{v} \rangle$ — Homogeneity

$\langle \underline{u}, \underline{u} \rangle \geq 0$ — Positivity, Note: $\langle \underline{u}, \underline{u} \rangle = 0$ if and only if $\underline{u} = \underline{0}$

An inner product space is a vector space equipped with an inner product. → literally any old vector space where the operation $\langle \underline{u}, \underline{v} \rangle$ is defined.

In an arbitrary inner product space:

norm: $\|\underline{v}\| = \sqrt{\langle \underline{v}, \underline{v} \rangle}$

distance: $d(\underline{u}, \underline{v}) = \|\underline{u} - \underline{v}\|$

## Examples of Inner Products (why are there so many???)

### Euclidean Inner Product on $\mathbb{R}^n$:

Given $\underline{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}$, $\underline{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$:

$\langle \underline{u}, \underline{v} \rangle = u_1v_1 + u_2v_2 + \dots + u_nv_n$

(dot product)

### Weighted Euclidean Inner Product on $\mathbb{R}^n$:

Given $\underline{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}$, $\underline{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$

and positive weights $w_1, \dots, w_n \in \mathbb{R}$:

$\langle \underline{u}, \underline{v} \rangle = w_1u_1v_1 + w_2u_2v_2 + \dots + w_nu_nv_n$

← Weights have to be positive to satisfy the positivity axiom for inner products.

### Matrix Inner Product on $\mathbb{R}^n$:

Given $\underline{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}$, $\underline{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$

and an invertible $n \times n$ matrix $A$:

$\langle \underline{u}, \underline{v} \rangle = A\underline{u} \cdot A\underline{v}$

→ The Euclidean and Weighted Euclidean inner products are both generalizations of the matrix inner product, with $A = \underbrace{\begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{pmatrix}}_{\text{Identity matrix}}$ and $A = \underbrace{\begin{pmatrix} \sqrt{w_1} & 0 & \cdots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & \sqrt{w_n} \end{pmatrix}}_{\mathrm{diag}(\sqrt{w_1}, \dots, \sqrt{w_n})}$ respectively

### Standard Inner Product on $\mathbb{P}_n$ → The space of polynomials of degree $\leq n$.

Given $\underline{p} = a_0 + a_1x + \dots + a_nx^n$,
$\underline{q} = b_0 + b_1x + \dots + b_nx^n$:

$\langle \underline{p}, \underline{q} \rangle = a_0b_0 + a_1b_1 + \dots + a_nb_n$

### Evaluation Inner Product on $\mathbb{P}_n$

Given $\underline{p} = p(x) \in \mathbb{P}_n$, $\underline{q} = q(x) \in \mathbb{P}_n$

and distinct sample points $x_0, x_1, \dots, x_n \in \mathbb{R}$:

$\langle \underline{p}, \underline{q} \rangle = p(x_0)q(x_0) + p(x_1)q(x_1) + \dots + p(x_n)q(x_n)$

(basically evaluate polynomials p and q at $x_0$ and multiply them together. Repeat for each x and add them all up)

Numerical example: Consider $\mathbb{P}_2$ with evaluation inner product at $x_0 = -2, x_1 = 0, x_2 = 1$

with $\underline{p} = x^2$, $\underline{q} = x+1$

$\langle \underline{p}, \underline{q} \rangle = (-2)^2(-2+1) + (0^2)(0+1) + (1^2)(1+1)$

$= -4 + 0 + 2 = -2$

$\|\underline{p}\| = \sqrt{\langle \underline{p}, \underline{p} \rangle} = \sqrt{16 + 0 + 1}$

$= \sqrt{17}$

Unit vector $\underline{p}' = \frac{1}{\|\underline{p}\|}\underline{p} = \frac{1}{\sqrt{17}}x^2$

## Inner Product on $C[a,b]$ → The space of all functions continuous on the interval $[a,b]$

Given $\underline{f} = f(x)$ and $\underline{g} = g(x)$ in $C[a,b]$:

$$\langle \underline{f}, \underline{g} \rangle = \int_a^b f(x)g(x)\,dx$$

## Inner Product on $\mathbb{C}^n$ → The space of vectors with complex entries.

(Note $\mathbb{R}$ is a subset of $\mathbb{C}$, if $\underline{u}, \underline{v}$ real then this is just the dot product).

Given $\underline{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}, \underline{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{C}^n$:

$$\langle \underline{u}, \underline{v} \rangle = \underline{u}^{\dagger} \underline{v}$$

Where $\underline{u}^{\dagger}$ is the Hermitian Conjugation of $\underline{u}$
→ basically just turn $\underline{u}$ into a row vector and take the complex conjugate of its entries, then matrix multiply this by $\underline{v}$.

this is mathematical bollocks for "change the signs of all the $i$ terms in one of them, then do normal dot product"

e.g if $\underline{u} = \begin{pmatrix} 7 \\ 3-i \\ 5+2i \end{pmatrix}$ and $\underline{v} = \begin{pmatrix} 6-i \\ i \\ 3+2i \end{pmatrix}$:

$\underline{u}^{\dagger} = \begin{pmatrix} 7 & 3+i & 5-2i \end{pmatrix}$

$\langle \underline{u}, \underline{v} \rangle = \begin{pmatrix} 7 & 3+i & 5-2i \end{pmatrix} \begin{pmatrix} 6-i \\ i \\ 3+2i \end{pmatrix}$

$= 7(6-i) + (3+i)i + (5-2i)(3+2i)$

$= 42 - 7i + 3i - 1 + 15 + 4i + 4$

$= 60$

$\underline{u} = \begin{pmatrix} 7 \\ 3-i \\ 5+2i \end{pmatrix}, \underline{v} = \begin{pmatrix} 6-i \\ i \\ 3+2i \end{pmatrix}$

$\langle \underline{u}, \underline{v} \rangle = \begin{pmatrix} 7 \\ 3+i \\ 5-2i \end{pmatrix} \cdot \begin{pmatrix} 6-i \\ i \\ 3+2i \end{pmatrix}$ $= \begin{pmatrix} 7 \\ 3-i \\ 5+2i \end{pmatrix} \cdot \begin{pmatrix} 6+i \\ -i \\ 3-2i \end{pmatrix} = \langle \underline{v}, \underline{u} \rangle$ → because of symmetry it does not matter whether we do this to $\underline{u}$ or $\underline{v}$.

$= 7(6-i) + (3+i)i + (5-2i)(3+2i)$

$= 60$

Finally! No more inner products :)

Note: Standard Inequalities:

$\|\underline{u}+\underline{v}\|^2 = \|\underline{u}\|^2 + \|\underline{v}\|^2$ for $\underline{u}, \underline{v}$ orthogonal — Pythagoras' Theorem

$|\langle \underline{u}, \underline{v} \rangle| \leq \|\underline{u}\| \|\underline{v}\|$ — Cauchy-Schwarz Theorem

$\|\underline{u}+\underline{v}\| \leq \|\underline{u}\| + \|\underline{v}\|$ — Triangle Inequality

## Orthogonality

Two vectors $\underline{u}$ and $\underline{v}$ in an inner product space are orthogonal if $\langle \underline{u}, \underline{v} \rangle = 0$

For example, $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are orthogonal under the Euclidean inner product in $\mathbb{R}^2$.

Given a subspace $W$ of an inner product space $V$, the subspace $W^{\perp} = \{\underline{x} \in V : \langle \underline{u}, \underline{x} \rangle = 0 \ \forall \underline{u} \in W\}$ is the Orthogonal Complement of $W$.

"Every vector in $W^{\perp}$ is orthogonal to every vector in $W$."

For example, given $V = \mathbb{R}^3$ with the dot product, and $W = \text{Span}\left\{\begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix}\right\}$:

$W^{\perp}$ is the solution space satisfying $\left\langle \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix}, \underline{x} \right\rangle = 0$ and $\left\langle \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix}, \underline{x} \right\rangle = 0$

Letting $\underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ this gives:

$$2x_1 + x_2 + 6x_3 = 0$$
$$3x_2 + x_3 = 0$$

Using Gauss-Jordan:

$$\left(\begin{array}{ccc|c} 2 & 1 & 6 & 0 \\ 0 & 3 & 1 & 0 \end{array}\right) \xrightarrow{\frac{1}{3}R_2} \left(\begin{array}{ccc|c} 2 & 1 & 6 & 0 \\ 0 & 1 & \frac{1}{3} & 0 \end{array}\right) \xrightarrow{R_1 - R_2} \left(\begin{array}{ccc|c} 2 & 0 & \frac{17}{3} & 0 \\ 0 & 1 & \frac{1}{3} & 0 \end{array}\right) \xrightarrow{\frac{1}{2}R_1} \left(\begin{array}{ccc|c} 1 & 0 & \frac{17}{6} & 0 \\ 0 & 1 & \frac{1}{3} & 0 \end{array}\right)$$

$x_1 + \frac{17}{6}x_3 = 0$

$x_2 + \frac{1}{3}x_3 = 0$

$x_3$ free variable → let $x_3 = \mu \in \mathbb{R}$

$x_1 = -\frac{17}{6}\mu$

$x_2 = -\frac{1}{3}\mu$

$\therefore \underline{u} = \mu \begin{pmatrix} -\frac{17}{6} \\ -\frac{1}{3} \\ 1 \end{pmatrix} = \frac{1}{6}\mu \begin{pmatrix} -17 \\ -2 \\ 6 \end{pmatrix}$

$\therefore W^{\perp} = \text{Span}\left\{ \begin{pmatrix} -17 \\ -2 \\ 6 \end{pmatrix} \right\}$

An orthogonal set is a set of vectors in an inner product space where all pairs of distinct vectors in the set are orthogonal
An orthonormal set is an orthogonal set consisting only of unit vectors.

Orthogonal/orthonormal sets of non-zero vectors are always linearly independent.
An orthogonal/orthonormal set that is also a basis is called an orthogonal/orthonormal basis

Theorem: If $S = \{\underline{v}_1, \ldots, \underline{v}_n\}$ is an orthogonal basis of an inner product space $V$, then for any vector $\underline{u} \in V$:

$$\underline{u} = \frac{\langle \underline{u}, \underline{v}_1 \rangle}{\|\underline{v}_1\|^2} \underline{v}_1 + \frac{\langle \underline{u}, \underline{v}_2 \rangle}{\|\underline{v}_2\|^2} \underline{v}_2 + \ldots + \frac{\langle \underline{u}, \underline{v}_n \rangle}{\|\underline{v}_n\|^2} \underline{v}_n$$

Proof: Since $\underline{u}$ is in the vector space $V$, it can be written as a linear combination of the basis vectors of $V$, i.e.

$\underline{u} = c_1\underline{v}_1 + c_2\underline{v}_2 + \ldots + c_n\underline{v}_n$ for some $c_1, \ldots, c_n \in \mathbb{R}$.

To find $c_1$, calculate $\langle \underline{u}, \underline{v}_1 \rangle = \langle c_1\underline{v}_1 + c_2\underline{v}_2 + \ldots + c_n\underline{v}_n, \underline{v}_1 \rangle$. Since $\underline{v}_1, \ldots, \underline{v}_n$ is an orthogonal basis, $\langle \underline{v}_1, \underline{v}_j \rangle = 0$ for all $j \neq 1$

This results in $\langle \underline{u}, \underline{v}_1 \rangle = \langle c_1\underline{v}_1, \underline{v}_1 \rangle$

$\downarrow$

by homogeneity axiom $\langle c_1\underline{v}_1, \underline{v}_1 \rangle = c_1 \langle \underline{v}_1, \underline{v}_1 \rangle$
$= c_1 \|\underline{v}_1\|^2$

Rearranging for $c_1$ gives $c_1 = \dfrac{\langle \underline{u}, \underline{v}_1 \rangle}{\|\underline{v}_1\|^2}$

Generalising this process for $c_i$ gives $c_i = \dfrac{\langle \underline{u}, \underline{v}_i \rangle}{\|\underline{v}_i\|^2}$

So $\underline{u} = c_1\underline{v}_1 + c_2\underline{v}_2 + \ldots + c_n\underline{v}_n$

$$= \frac{\langle \underline{u}, \underline{v}_1 \rangle}{\|\underline{v}_1\|^2} \underline{v}_1 + \frac{\langle \underline{u}, \underline{v}_2 \rangle}{\|\underline{v}_2\|^2} \underline{v}_2 + \ldots + \frac{\langle \underline{u}, \underline{v}_n \rangle}{\|\underline{v}_n\|^2} \underline{v}_n$$

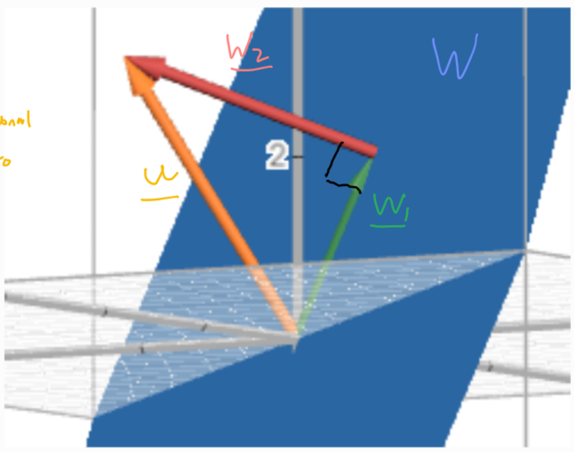
## Orthogonal Projections

Projection Theorem: Let $W$ be a subspace of a finite-dimensional inner product space $V$.
Then, every vector $\underline{u}$ in $V$ can be expressed as $\underline{u} = \underline{w}_1 + \underline{w}_2$, where $\underline{w}_1 \in W$, $\underline{w}_2 \in W^{\perp}$.
Here, $\underline{w}_1$ is the orthogonal projection of $\underline{u}$ onto $W$, i.e. the closest possible approximation of $\underline{u}$ within $W$.

It makes sense for $\underline{w}_2$ to be in $W^{\perp}$ here (given that $\underline{w}_1$ is the orthogonal projection). Consider an example where $V$ is a 3 dimensional inner product space, with $W$ a 2-dimensional subspace and $\underline{u}$ a vector in $V$. Thinking geometrically, the shortest distance between a point (representing $\underline{u}$) in 3D space and a 2D plane (representing $W$) in 3D space is the perpendicular distance. In the context of orthogonal projections, we want to find $\underline{w}_1$ which is the closest approximation of $\underline{u}$ in $W$, meaning we want the point on $W$ with the shortest distance to $\underline{u}$ – which is the perpendicular distance. Therefore, $\underline{w}_2$ $(= \underline{u} - \underline{w}_1)$ will be perpendicular, or orthogonal, to $W$, and hence is in $W^{\perp}$.

$\downarrow$

This applies to any finite dimensional spaces, but can be somewhat hard to visualise more than 3 dimensions :)




(desmos 3D is fucking awesome)

Notation: $\underline{w}_1 = \text{proj}_W \underline{u}$
$\underline{w}_2 = \text{proj}_{W^\perp} \underline{u}$

Computing $\underline{w}_1$ and $\underline{w}_2$:

Given $\{\underline{v}_1, \underline{v}_2, \dots, \underline{v}_n\}$ is an orthogonal basis for an n-dimensional subspace $W$, which is a subspace of an r-dimensional inner product space $V$, and $\underline{u}$ is a vector in $V$, we can calculate $\text{proj}_W \underline{u}$ $(= \underline{w}_1)$

Since $\text{proj}_W \underline{u}$ is in $W$, we can rewrite it as a linear combination of the basis vectors of $W$:

① $\text{proj}_W \underline{u} = c_1 \underline{v}_1 + c_2 \underline{v}_2 + \dots + c_n \underline{v}_n$, $c_1, \dots, c_n \in \mathbb{R}$

Now calculate the inner product of $\underline{u}$ and $\underline{v}_1$:

② $\langle \underline{u}, \underline{v}_1 \rangle = \langle \text{proj}_W \underline{u} + \text{proj}_{W^\perp} \underline{u}, \underline{v}_1 \rangle = \langle \text{proj}_W \underline{u}, \underline{v}_1 \rangle + \underbrace{\langle \text{proj}_{W^\perp} \underline{u}, \underline{v}_1 \rangle}$ $= \langle \text{proj}_W \underline{u}, \underline{v}_1 \rangle$

by additivity axiom

Since $\underline{v}_1 \in W$ and $\text{proj}_{W^\perp} \underline{u} \in W^\perp$, these are orthogonal so this inner product is zero

Substituting ① into ② gives:

$\langle \underline{u}, \underline{v}_1 \rangle = \langle c_1 \underline{v}_1 + c_2 \underline{v}_2 + \dots + c_n \underline{v}_n, \underline{v}_1 \rangle$

Since $\underline{v}_1, \dots, \underline{v}_n$ is an orthogonal basis, $\langle \underline{v}_1, \underline{v}_j \rangle = 0$ for all $j \neq 1$, this results in:

$\langle \underline{u}, \underline{v}_1 \rangle = \langle c_1 \underline{v}_1, \underline{v}_1 \rangle$

Applying the homogeneity axiom gives:

$\langle \underline{u}, \underline{v}_1 \rangle = c_1 \langle \underline{v}_1, \underline{v}_1 \rangle = c_1 \|\underline{v}_1\|^2$

$\Rightarrow c_1 = \dfrac{\langle \underline{u}, \underline{v}_1 \rangle}{\|\underline{v}_1\|^2}$

Repeating this process for each $c_1, \dots, c_n$ gives:

$\text{proj}_W \underline{u} = c_1 \underline{v}_1 + c_2 \underline{v}_2 + \dots + c_n \underline{v}_n$

$$\boxed{\text{proj}_W \underline{u} = \frac{\langle \underline{u}, \underline{v}_1 \rangle}{\|\underline{v}_1\|^2} \underline{v}_1 + \frac{\langle \underline{u}, \underline{v}_2 \rangle}{\|\underline{v}_2\|^2} \underline{v}_2 + \dots + \frac{\langle \underline{u}, \underline{v}_n \rangle}{\|\underline{v}_n\|^2} \underline{v}_n}$$

From this, calculating $\text{proj}_{W^\perp} \underline{u}$ is simple:

$\text{proj}_{W^\perp} \underline{u} = \underline{u} - \text{proj}_W \underline{u}$

## Example

Consider $\mathbb{R}^3$ with the Weighted Euclidean Inner Product with weights $w_1 = 2, w_2 = 3, w_3 = 3$. Let $W$ be the subspace formed by $\text{Span}\left\{\begin{pmatrix}0\\0\\1\end{pmatrix}, \begin{pmatrix}-2\\3\\0\end{pmatrix}\right\}$.

Express $\underline{u} = \begin{pmatrix}5\\3\\2\end{pmatrix}$ in the form $\underline{u} = \underline{w}_1 + \underline{w}_2$, where $\underline{w}_1 \in W$ and $\underline{w}_2 \in W^\perp$

Using the formula:

$\underline{w}_1 = \text{proj}_W \underline{u} = \dfrac{\langle \underline{u}, \underline{v}_1 \rangle}{\|\underline{v}_1\|^2} \underline{v}_1 + \dfrac{\langle \underline{u}, \underline{v}_2 \rangle}{\|\underline{v}_2\|^2} \underline{v}_2$

$\underline{u} = \begin{pmatrix}5\\3\\2\end{pmatrix}, \underbrace{\underline{v}_1 = \begin{pmatrix}0\\0\\1\end{pmatrix}, \underline{v}_2 = \begin{pmatrix}-2\\3\\0\end{pmatrix}}$

orthogonal basis :)

$= \dfrac{3 \times 2 \times 1}{3} \begin{pmatrix}0\\0\\1\end{pmatrix} + \dfrac{2 \times (-2) \times 5 + 3 \times 3 \times 3}{2 \times (-2) \times (-2) + 3 \times 3 \times 3} \begin{pmatrix}-2\\3\\0\end{pmatrix}$

Note: $\|\underline{v}\| = \sqrt{\langle \underline{v}_1, \underline{v}_1 \rangle}$ is dependent on the inner product used, it is only Pythagoras when using the Euclidean inner product.

$= 2 \begin{pmatrix}0\\0\\1\end{pmatrix} + \dfrac{7}{35} \begin{pmatrix}-2\\3\\0\end{pmatrix} = \begin{pmatrix}0\\0\\2\end{pmatrix} + \begin{pmatrix}-2/5\\3/5\\0\end{pmatrix} = \begin{pmatrix}-2/5\\3/5\\2\end{pmatrix}$

$$\underline{w}_2 = \underline{u} - \underline{w}_1 = \begin{pmatrix} 5 \\ 3 \\ 2 \end{pmatrix} - \begin{pmatrix} -2/5 \\ 3/5 \\ 2 \end{pmatrix} = \begin{pmatrix} 27/5 \\ 12/5 \\ 0 \end{pmatrix}$$

$$\therefore \underline{u} = \begin{pmatrix} -2/5 \\ 3/5 \\ 2 \end{pmatrix} + \begin{pmatrix} 27/5 \\ 12/5 \\ 0 \end{pmatrix}$$

## Gram-Schmidt Orthogonalisation Process

Every non-zero finite dimensional inner product space has an orthonormal basis.

Consider an inner product space $V$, with a subspace $W$, and let $\{\underline{u}_1, \dots, \underline{u}_n\}$ be a non-orthogonal basis of $W$.

The Gram-Schmidt process allows us to find an orthogonal basis $\{\underline{v}_1, \dots, \underline{v}_n\}$ of $W$.

It does this by considering the subspaces $W_r = \text{Span}(\underline{u}_1, \dots, \underline{u}_r)$ for $r=1$ through to $n$, building up an orthogonal basis for $W$.

Note: $W_1 \subseteq W_2 \subseteq \cdots \subseteq W_n$

Once we have found the orthogonal basis, we can normalise (unit vector-ify) them to get an orthonormal basis.

The process: (Note: $\{\underline{u}_1, \dots, \underline{u}_n\}$ is the given non-orthogonal basis, $\{\underline{v}_1, \dots, \underline{v}_n\}$ is the orthogonal basis we want to find)

① Consider $W_1 = \text{Span}(\underline{u}_1)$

$\underline{u}_1$ is obviously an orthogonal basis of $W_1$, so set $\underline{v}_1 = \underline{u}_1$

② Consider $W_2 = \text{Span}(\underline{u}_1, \underline{u}_2)$

We already have an orthogonal basis for $W_1 = \text{Span}(\underline{u}_1)$, which is $\underline{v}_1$. This means we can access any vector in $W_1$ with some linear combination of this basis. This means that the component of $\underline{u}_2$ which lies within $W_1$, i.e. $\text{proj}_{W_1}\underline{u}_2$ can be expressed in terms of the basis of $W_1$, so we can subtract $\text{proj}_{W_1}\underline{u}_2$ from $\underline{u}_2$ before adding it to the basis, without affecting the span of vectors the basis represents. Since $\underline{u}_2 = \text{proj}_{W_1}\underline{u}_2 + \text{proj}_{W_1^\perp}\underline{u}_2$, the subtraction $\underline{u}_2 - \text{proj}_{W_1}\underline{u}_2 = \text{proj}_{W_1^\perp}\underline{u}_2$ gives us the component of $\underline{u}_2$ in $W_1^\perp$, which is orthogonal to $W_1$ and its basis. So, setting:

$$\underline{v}_2 = \underline{u}_2 - \text{proj}_{W_1}\underline{u}_2$$
$$= \underline{u}_2 - \frac{\langle \underline{u}_2, \underline{v}_1 \rangle}{\|\underline{v}_1\|^2}\underline{v}_1$$

gives us an orthogonal basis $\{\underline{v}_1, \underline{v}_2\}$ of $W_2$, with the same span as $\{\underline{u}_1, \underline{u}_2\}$

③ Consider $W_3 = \text{Span}(\underline{u}_1, \underline{u}_2, \underline{u}_3)$

In step ② we found an orthogonal basis for $W_2 = \text{Span}(\underline{u}_1, \underline{u}_2)$, namely $\{\underline{v}_1, \underline{v}_2\}$, so we can rewrite $W_3 = \text{Span}(\underline{v}_1, \underline{v}_2, \underline{u}_3)$

Since $\underline{v}_1$ and $\underline{v}_2$ are already orthogonal, we only need to focus on $\underline{u}_3$. Like before, we need to use orthogonal projections to find the component of $\underline{u}_3$ that is orthogonal to $W_2$, i.e. we need $\text{proj}_{W_2^\perp}\underline{u}_3 = \underline{v}_3$:

$$\underline{v}_3 = \underline{u}_3 - \text{proj}_{W_2}\underline{u}_3$$
$$= \underline{u}_3 - \frac{\langle \underline{u}_3, \underline{v}_1 \rangle}{\|\underline{v}_1\|^2}\underline{v}_1 - \frac{\langle \underline{u}_3, \underline{v}_2 \rangle}{\|\underline{v}_2\|^2}\underline{v}_2$$

Note: Since applying the Projection Theorem to $\underline{u}_3$ gives

$$\underline{u}_3 = \text{proj}_{W_2}\underline{u}_3 + \text{proj}_{W_2^\perp}\underline{u}_3$$
$$= \text{proj}_{W_2}\underline{u}_3 + \underline{v}_3$$

and $\text{proj}_{W_2}\underline{u}_3$ can be rewritten as a linear combination of the orthogonal basis vectors of $W_2$:

$$\text{proj}_{W_2}\underline{u}_3 \in W_2 \Rightarrow \text{proj}_{W_2}\underline{u}_3 = a\underline{v}_1 + b\underline{v}_2 \qquad a, b \in \mathbb{R}$$

Then

$$\text{Span}(\underline{v}_1, \underline{v}_2, \underline{v}_3) = \text{Span}(\underline{u}_1, \underline{u}_2, \underline{u}_3) = W_3$$

So the process works ☺

Generalising Step ③ for $W_r$, $1 < r \leq n$, gives:

(r)

$$\boxed{\begin{aligned}\underline{v}_r &= \text{proj}_{W_{r-1}^{\perp}} \underline{u}_r \\ &= \underline{u}_r - \text{proj}_{W_r} \underline{u}_r \\ &= \underline{u}_r - \frac{\langle \underline{u}_r, \underline{v}_1 \rangle}{\|\underline{v}_1\|^2} \underline{v}_1 - \frac{\langle \underline{u}_r, \underline{v}_2 \rangle}{\|\underline{v}_2\|^2} \underline{v}_2 - \cdots - \frac{\langle \underline{u}_r, \underline{v}_{r-1} \rangle}{\|\underline{v}_{r-1}\|^2} \underline{v}_{r-1}\end{aligned}}$$

(repeat until all of $\{\underline{v}_1, \ldots, \underline{v}_n\}$ have been found)

If an orthonormal basis is needed, divide each vector by its norm: $\left\{ \frac{\underline{v}_1}{\|\underline{v}_1\|}, \ldots, \frac{\underline{v}_n}{\|\underline{v}_n\|} \right\}$

## Example in $\mathbb{R}^3$

Consider $\mathbb{R}^3$ with the Euclidean inner product. Find an orthonormal basis of $W = \text{Span}\left\{ \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}$

$$\underline{v}_1 = \underline{u}_1 = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

$$\underline{v}_2 = \underline{u}_2 - \text{proj}_{W_1} \underline{u}_2$$

$$= \underline{u}_2 - \frac{\langle \underline{u}_2, \underline{v}_1 \rangle}{\|\underline{v}_1\|^2} \underline{v}_1$$

$$= \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} - \frac{2+6+1}{14} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} - \frac{9}{14} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} -4/14 \\ -1/14 \\ 5/14 \end{pmatrix}$$

Since fractions are ugly and its going to be normalized later:

$$\underline{v}_2 = \begin{pmatrix} -4 \\ 1 \\ 5 \end{pmatrix}$$

$$\underline{v}_3 = \underline{u}_3 - \text{proj}_{W_2} \underline{u}_3$$

$$= \underline{u}_3 - \frac{\langle \underline{u}_3, \underline{v}_1 \rangle}{\|\underline{v}_1\|^2} \underline{v}_1 - \frac{\langle \underline{u}_3, \underline{v}_2 \rangle}{\|\underline{v}_2\|^2} \underline{v}_2$$

$$= \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \frac{2}{14} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} + \frac{4}{42} \begin{pmatrix} -4 \\ -1 \\ 5 \end{pmatrix} = \begin{pmatrix} 21/21 \\ 0 \\ 0 \end{pmatrix} - \frac{3}{21} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} + \frac{2}{21} \begin{pmatrix} -4 \\ 1 \\ 5 \end{pmatrix}$$

$$= \begin{pmatrix} 7/21 \\ -7/21 \\ 7/21 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

$\therefore$ orthogonal basis $\begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} -4 \\ 1 \\ 5 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

Orthonormal basis $\begin{pmatrix} 2/\sqrt{14} \\ 3/\sqrt{14} \\ 1/\sqrt{14} \end{pmatrix}, \begin{pmatrix} -4/\sqrt{42} \\ 1/\sqrt{42} \\ 5/\sqrt{42} \end{pmatrix}, \begin{pmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}$

## Example in $C[-1,1]$ → The set of functions continuous in $[-1, 1]$

Consider the space $C[-1,1]$. Find an orthogonal basis of $W = \text{Span}(\underline{u}_1, \underline{u}_2, \underline{u}_3)$ with $\underline{u}_1 = 1$, $\underline{u}_2 = x$, $\underline{u}_3 = x^2$

let $\underline{v}_1 = \underline{u}_1 = 1$

$$\underline{v}_2 = \underline{u}_2 - \text{proj}_W \underline{u}_2$$

$$= \underline{u}_2 - \frac{\langle \underline{u}_2, \underline{v}_1 \rangle}{\|\underline{v}_1\|^2} \underline{v}_1$$

$$= x - \frac{\int_{-1}^{1} x \, dx}{\int_{-1}^{1} 1 \, dx} \times 1$$

$\longrightarrow$ in $C[a,b]$, $\langle \underline{u}, \underline{v} \rangle = \int_a^b f(x) g(x) dx$, where $\underline{u} = f(x)$, $\underline{v} = g(x)$

$\|\underline{v}\| = \langle \underline{v}, \underline{v} \rangle = \int_a^b g(x)^2 dx$

$$= x - 0 = x$$

$$\underline{v}_3 = \underline{u}_3 - \frac{\langle \underline{u}_3, \underline{v}_1 \rangle}{\|\underline{v}_1\|^2} \underline{v}_1 - \frac{\langle \underline{u}_3, \underline{v}_2 \rangle}{\|\underline{v}_2\|} \underline{v}_2$$

$$= x^2 - \frac{\int_{-1}^{1} x^2 dx}{\int_{-1}^{1} 1 \, dx} \times 1 - \frac{\int_{-1}^{1} x^3 dx}{\int_{-1}^{1} x^2 dx} x$$

$$= x^2 - \frac{2/3}{2}$$

$$= x^2 - 1/3$$

$\therefore$ orthogonal basis $\{1, x, x^2 - \frac{1}{3}\}$

# QR Decomposition

Any $m \times n$ matrix $A$ with linearly independent columns can be factored as $A = QR$, with $Q$ having orthonormal columns and $R$ being invertible and upper triangular. We can decompose $A$ into this form using Gram-Schmidt.

Let $A$ be an $m \times n$ matrix with linearly independent columns $\underline{u}_1, \dots, \underline{u}_n$

Let $\{\underline{v}_1, \dots, \underline{v}_n\}$ be the orthogonal set obtained by applying Gram-Schmidt to $\underline{u}_1, \dots, \underline{u}_n$, and $\underline{q}_r = \frac{\underline{v}_r}{\|\underline{v}_r\|}$ from $r = 1, \dots, n$.

i.e.:

$$\underline{v}_1 = \underline{u}_1$$

$$\underline{q}_1 = \frac{\underline{u}_1}{\|\underline{v}_1\|}$$

$$\underline{v}_r = \underline{u}_r - \frac{\langle \underline{u}_1, \underline{q}_1 \rangle}{\|\underline{q}_1\|^2} \underline{q}_1 - \frac{\langle \underline{u}_2, \underline{q}_2 \rangle}{\|\underline{q}_2\|^2} \underline{q}_2 - \dots - \frac{\langle \underline{u}_r, \underline{q}_{r-1} \rangle}{\|\underline{q}_{r-1}\|^2} \underline{q}_{r-1}$$

Since $\underline{q}_1, \dots, \underline{q}_n$ orthonormal, $\|\underline{q}_1\|, \dots, \|\underline{q}_n\| = 1$

$$\underline{v}_r = \underline{u}_r - \langle \underline{u}_1, \underline{q}_1 \rangle \underline{q}_1 - \langle \underline{u}_2, \underline{q}_2 \rangle \underline{q}_2 - \dots - \langle \underline{u}_r, \underline{q}_{r-1} \rangle \underline{q}_{r-1}$$

$$\underline{q}_r = \frac{\underline{v}_r}{\|\underline{v}_r\|}$$

This results in a matrix $Q$ with orthonormal columns $\underline{q}_1, \dots, \underline{q}_n$. Since $A = QR$, to find $R$ we need to work out what to multiply $Q$ by to get $A$. Since $A = \begin{pmatrix} | & & | \\ \underline{u}_1 & \cdots & \underline{u}_n \\ | & & | \end{pmatrix}$, we can rearrange the above equations to get each $\underline{u}_r$ in terms of $\underline{q}_1, \dots, \underline{q}_{r-1}$

$$\underline{u}_1 = \|\underline{v}_1\| \underline{q}_1 \left( + 0\underline{q}_2 + 0\underline{q}_3 + \dots + 0\underline{q}_n \right)$$

$\longrightarrow$ $\underline{u}_1$ is orthogonal to $\underline{q}_2, \dots, \underline{q}_n$ since $\underline{q}_1$ is an orthonormal basis for $\underline{u}_1$ by Gram-Schmidt, so $\underline{u}_1$ is a linear combination of $\underline{q}_1$ with $0 \times \underline{q}_2, \dots, \underline{q}_n$.

$\downarrow$

Generalizing to $\underline{u}_r$: $\underline{q}_1, \dots, \underline{q}_r$ is a basis for $\underline{u}_r$ so none of $\underline{q}_{r+1}, \dots, \underline{q}_n$ are added to get $\underline{u}_r$

$$= \langle \underline{u}_1, \underline{q}_1 \rangle \underline{q}_1$$

$\rightarrow$ Note: $\|\underline{v}_r\| \underline{q}_r = \langle \underline{u}_r, \underline{q}_r \rangle \underline{q}_r$

$$\underline{u}_2 = \underline{v}_2 + \langle \underline{u}_1, \underline{q}_1 \rangle \underline{q}_1$$

$$= \|\underline{v}_2\| \underline{q}_2 + \langle \underline{u}_1, \underline{q}_1 \rangle \underline{q}_1$$

$$= \langle \underline{u}_1, \underline{q}_1 \rangle \underline{q}_1 + \langle \underline{u}_2, \underline{q}_2 \rangle \underline{q}_2 \left( + 0\underline{q}_3 + \dots + 0\underline{q}_n \right)$$

Generalising for $\underline{u}_r$ gives:

$$\underline{u}_r = \langle \underline{u}_r, \underline{q}_1 \rangle \underline{q}_1 + \langle \underline{u}_2, \underline{q}_2 \rangle \underline{q}_2 + \dots + \langle \underline{u}_r, \underline{q}_r \rangle \underline{q}_r \left( + 0\underline{q}_{r+1} + \dots + 0\underline{q}_n \right)$$

Looking back at the matrix multiplication $A = QR$, each coefficient of $\underline{q}_i$ in the above equation for each $\underline{u}_r$ forms the entry in the $i^{th}$ row and $r^{th}$ column of $R$, i.e.:

$$R = \begin{pmatrix} \langle \underline{u}_1, \underline{q}_1 \rangle & \langle \underline{u}_2, \underline{q}_1 \rangle & \cdots & \langle \underline{u}_n, \underline{q}_1 \rangle \\ 0 & \langle \underline{u}_2, \underline{q}_2 \rangle & \cdots & \langle \underline{u}_n, \underline{q}_2 \rangle \\ \vdots & 0 & & \vdots \\ \vdots & \vdots & & \\ 0 & 0 & & \langle \underline{u}_n, \underline{q}_n \rangle \end{pmatrix}$$

which is upper triangular.

Writing $A$ in the form $A = QR$ is known as the QR Decomposition of $A$.

## Least Squares Solutions of Inconsistent Linear Systems

Least squares problem: given a linear system $A\underline{x} = \underline{b}$ with $m$ equations and $n$ variables, we want to find a vector $\underline{x}$ to minimise $\| \underline{b} - A\underline{x} \|$

Consider the vector subspace $W = \{ A\underline{x} \mid \underline{x} \in \mathbb{R}^n \}$, which is the column space of $A$, i.e. the span of the columns of $A$. This subspace contains all possible values of $A\underline{x}$.

We want to minimise $\|\underline{b} - A\underline{x}\|$, i.e. we want the best approximation to $\underline{b}$ within $W$. This is $\text{proj}_W \underline{b}$.

This means that the least squares solutions of $A\underline{x} = \underline{b}$ are the exact solutions to $A\underline{x} = \text{proj}_W \underline{b}$. $\rightarrow$ We could technically compute $\text{proj}_W \underline{b}$ and solve at this stage but Karl doesn't do it like that

Since $\underline{b} = \text{proj}_W \underline{b} + \text{proj}_{W^\perp} \underline{b}$ is unique, we get $A\underline{x} = \underline{b} - \text{proj}_{W^\perp} \underline{b}$, i.e. $\underline{b} - A\underline{x} \in W^\perp$. This means that the inner product of any column $\underline{a}_i$ of $A$ with the vector $\underline{b} - A\underline{x}$ would yield $0$. For the Euclidean inner product, this is equivalent to matrix multiplying the transpose of $\underline{a}_i$ with $\underline{b} - A\underline{x}$. This means that $A^T(\underline{b} - A\underline{x}) = \underline{0}$, or equivalently:

$$\boxed{A^T A \underline{x} = A^T \underline{b}}$$

This is the normal equation associated with $A\underline{x} = \underline{b}$. Solving for $\underline{x}$ using Gauss-Jordan will yield the least squares solution.

### Example

Find least squares solutions for:

$$x - y = 4$$
$$3x + 2y = 1$$
$$-2x + 4y = 3$$

Using the Euclidean inner product

$$A\underline{x} = \underline{b}$$

$$\begin{pmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix}$$

Left-multiplying by $A^T$ gives:

$$\begin{pmatrix} 1 & 3 & -2 \\ -1 & 2 & 4 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 3 & -2 \\ -1 & 2 & 4 \end{pmatrix} \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix}$$

Multiplying out:

$$\begin{pmatrix} 14 & -3 \\ -3 & 21 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 10 \end{pmatrix} \to \text{a nice system}$$

Gauss-Jordan:

$$\left(\begin{array}{cc|c} 14 & -3 & 1 \\ -3 & 21 & 10 \end{array}\right) \xrightarrow[\text{Swap } R_1 \text{ and } R_2]{-1/3 \times R_2} \left(\begin{array}{cc|c} 1 & -7 & -\frac{10}{3} \\ 14 & -3 & 1 \end{array}\right) \xrightarrow{R_2 - 14R_1} \left(\begin{array}{cc|c} 1 & -7 & -\frac{10}{3} \\ 0 & 95 & \frac{143}{3} \end{array}\right) \xrightarrow[R_1 + 7R_2]{\frac{1}{95}R_2} \left(\begin{array}{cc|c} 1 & 0 & \frac{17}{95} \\ 0 & 1 & \frac{143}{285} \end{array}\right)$$

ugly fractions

$\therefore$ least squares solution $x = \frac{17}{95}$, $y = \frac{143}{285}$

Note: $\underline{b} - A\underline{x}$ is called the error vector.

$\|\underline{b} - A\underline{x}\|$ is called the error.

# Linear Regression

Least Squares Solutions can be used to fit a line or curve to a set of data points on a graph.

E.g. to fit a straight line $y = a + bx$ to a series of data points $(x_1, y_1), (x_2, y_2), \ldots$, you can find the least squares solutions to

$$\begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \end{pmatrix}$$

Theorem: Any $m \times n$ matrix $A$ has linearly independent columns if and only if $(A^TA)^{-1}$ exists.

From normal equation $A^TA\underline{x} = A^T\underline{b}$ and above theorem, a unique least squares solution $\underline{x} = (A^TA)^{-1}A^T\underline{b}$ only exists if $A$ has linearly independent columns.

Assuming $A$ does have linearly independent columns, we can use the QR decomposition of $A$ to compute $\underline{x}$:

$$\begin{aligned}
&\text{Let } A = QR \\
&\underline{x} = ((QR)^T QR)^{-1}(QR)^T\underline{b} && \to (AB)^T = B^TA^T \\
&\quad = (R^TQ^TQR)^{-1}R^TQ^T\underline{b} \\
&\quad = (R^TIR)^{-1}R^TQ^T\underline{b} && \to Q^TQ = I \text{ since } Q \text{ has orthonormal column vectors and } Q^T \\
&\quad = (R^TR)^{-1}R^TQ^T\underline{b} \\
&\quad = R^{-1}(R^T)^{-1}R^TQ^T\underline{b} && \to (AB)^{-1} = B^{-1}A^{-1} \\
&\quad = R^{-1}Q^T\underline{b}
\end{aligned}$$

This yields:

$$\boxed{\begin{gathered} \underline{x} = R^{-1}Q^T\underline{b} \\ \text{or} \\ R\underline{x} = Q^T\underline{b} \end{gathered}}$$

## Example

Given the data points $(-2,1), (-1,-2), (-3,6), (0,-6), (-2,0), (-1,-3)$, weighted as $(4,4,3,3,1,1)$, fit the data to a curve of the form $y = a + bx + cx^2$.

Form linear system $A\underline{x} = \underline{y}$

Since we are trying to find $a, b, c$, set $\underline{x} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$. Substitute each data point into $y = a + bx + cx^2$ and record the coefficients of $a$, $b$ and $c$ as rows of the matrix $A$, with each value of $y$ being added to $\underline{y}$.

Doing this with each point gives:

$$\begin{pmatrix} \boxed{1 & -2 & 4} \\ 1 & -1 & 1 \\ 1 & -3 & 9 \\ 1 & 0 & 0 \\ 1 & -2 & 4 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} \boxed{1} \\ -2 \\ 6 \\ -6 \\ 0 \\ -3 \end{pmatrix}$$

E.g. $(-2, 1)$

$y = a + bx + cx^2$

$\rightarrow \; a - 2b + 4c = 1$

Coefficients $(1 \;\; -2 \;\; 4)$

$y = 1$

Need QR decomposition of $A$

Gram-Schmidt to find $Q$ – need to orthogonalise columns

$$\underline{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

Note: we are using weighted Euclidean inner product here

Weights: $(4,4,3,3,1,1)$

$$\underline{v}_2 = \underline{u}_2 - \frac{\langle \underline{u}_2, \underline{v}_1 \rangle}{\langle \underline{v}_1, \underline{v}_1 \rangle} \underline{v}_1 = \begin{pmatrix} -2 \\ -1 \\ -3 \\ 0 \\ -2 \\ -1 \end{pmatrix} - \frac{-8-4-9+0-2-1}{4+4+3+3+1+1} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ -3 \\ 0 \\ -2 \\ -1 \end{pmatrix} + \frac{3}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1/2 \\ 1/2 \\ -3/2 \\ 3/2 \\ -1/2 \\ 1/2 \end{pmatrix}$$

Oh god...

$$\underline{v}_3 = \underline{u}_3 - \frac{\langle \underline{u}_3, \underline{v}_1 \rangle}{\langle \underline{v}_1, \underline{v}_1 \rangle} \underline{v}_1 - \frac{\langle \underline{u}_3, \underline{v}_2 \rangle}{\langle \underline{v}_2, \underline{v}_2 \rangle} \underline{v}_2 = \begin{pmatrix} 4 \\ 1 \\ 9 \\ 0 \\ 4 \\ 1 \end{pmatrix} - \frac{16+4+27+0+4+1}{4+4+3+3+1+1} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} - \frac{-8+2-\frac{81}{2}+0-2+\frac{1}{2}}{1+1+\frac{27}{4}+\frac{27}{4}+\frac{1}{4}+\frac{1}{4}} \begin{pmatrix} -1/2 \\ 1/2 \\ -3/2 \\ 3/2 \\ -1/2 \\ 1/2 \end{pmatrix}$$

$$= \begin{pmatrix} 4 \\ 1 \\ 9 \\ 0 \\ 4 \\ 1 \end{pmatrix} - \frac{13}{4} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} -1/2 \\ 1/2 \\ -3/2 \\ 3/2 \\ -1/2 \\ 1/2 \end{pmatrix}$$

$$= \begin{pmatrix} -3/4 \\ -3/4 \\ 5/4 \\ 5/4 \\ -3/4 \\ -3/4 \end{pmatrix}$$

Normalise each $\underline{v}$:

$$\underline{q}_1 = \frac{1}{\langle \underline{v}_1, \underline{v}_1 \rangle} \underline{v}_1 = \frac{1}{\sqrt{4+4+3+3+1+1}} \underline{v}_1 = \frac{1}{4} \underline{v}_1$$

Need to use weighted Euclidean inner product when normalising ($\|\underline{v}_i\| = \langle \underline{v}_i, \underline{v}_i \rangle$)

$$\underline{q}_2 = \frac{1}{\langle \underline{v}_2, \underline{v}_2 \rangle} \underline{v}_2 = \frac{1}{\sqrt{16}} \underline{v}_2 = \frac{1}{4} \underline{v}_2$$

$$\underline{q}_3 = \frac{1}{\langle \underline{v}_3, \underline{v}_3 \rangle} \underline{v}_3 = \frac{1}{\sqrt{15}} \underline{v}_3$$

$$\therefore Q = \begin{pmatrix} \frac14 & -\frac18 & \frac{-3}{4\sqrt{15}} \\ \frac14 & \frac18 & \frac{-3}{4\sqrt{15}} \\ \frac14 & -\frac38 & \frac{5}{4\sqrt{15}} \\ \frac14 & \frac38 & \frac{5}{4\sqrt{15}} \\ \frac14 & -\frac18 & \frac{-3}{4\sqrt{15}} \\ \frac14 & \frac18 & \frac{-3}{4\sqrt{15}} \end{pmatrix}$$

and we still need to find $R$....

For reference: $\begin{pmatrix} \underline{u}_1 & \underline{u}_2 & \underline{u}_3 \end{pmatrix} = \begin{pmatrix} 1 & -2 & 4 \\ 1 & -1 & 1 \\ 1 & -3 & 9 \\ 1 & 0 & 0 \\ 1 & -2 & 4 \\ 1 & -1 & 1 \end{pmatrix}$

weights $(4, 4, 3, 3, 1, 1)$

again, need to use weights here !!

$$R = \begin{pmatrix} \langle \underline{u}_1, \underline{q}_1 \rangle & \langle \underline{u}_2, \underline{q}_1 \rangle & \langle \underline{u}_3, \underline{q}_1 \rangle \\ 0 & \langle \underline{u}_2, \underline{q}_2 \rangle & \langle \underline{u}_3, \underline{q}_2 \rangle \\ 0 & 0 & \langle \underline{u}_3, \underline{q}_3 \rangle \end{pmatrix} = \begin{pmatrix} 4 & -6 & 13 \\ 0 & 4 & -12 \\ 0 & 0 & \sqrt{15} \end{pmatrix}$$

Still not done... we actually need $\underline{x} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$

Using normal equation $R\underline{x} = Q^T \underline{y}$ :

$$\underline{x} = R^{-1} Q^T \underline{y}$$

Need to find $R^{-1}$:

$$\left(\begin{array}{ccc|ccc} 4 & -6 & 13 & 1 & 0 & 0 \\ 0 & 4 & -12 & 0 & 1 & 0 \\ 0 & 0 & \sqrt{15} & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 4 & -6 & 13 & 1 & 0 & 0 \\ 0 & 4 & -12 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{\sqrt{15}} \end{array}\right) \to \left(\begin{array}{ccc|ccc} 4 & -6 & 0 & 1 & 0 & -\frac{13}{\sqrt{15}} \\ 0 & 1 & 0 & 0 & \frac14 & \frac{3}{\sqrt{15}} \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{\sqrt{15}} \end{array}\right)$$

$$\to \left(\begin{array}{ccc|ccc} 1 & -\frac32 & 0 & \frac14 & 0 & \frac{-13}{4\sqrt{15}} \\ 0 & 1 & 0 & 0 & \frac14 & \frac{3}{\sqrt{15}} \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{\sqrt{15}} \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac14 & \frac38 & \frac{5}{4\sqrt{15}} \\ 0 & 1 & 0 & 0 & \frac14 & \frac{3}{\sqrt{15}} \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{\sqrt{15}} \end{array}\right)$$

$$\therefore R^{-1} = \begin{pmatrix} \frac14 & \frac38 & \frac{5}{4\sqrt{15}} \\ 0 & \frac14 & \frac{3}{\sqrt{15}} \\ 0 & 0 & \frac{1}{\sqrt{15}} \end{pmatrix}$$

$$\underline{x} = \begin{pmatrix} \frac14 & \frac38 & \frac{5}{4\sqrt{15}} \\ 0 & \frac14 & \frac{3}{\sqrt{15}} \\ 0 & 0 & \frac{1}{\sqrt{15}} \end{pmatrix} \begin{pmatrix} \frac14 & \frac14 & \frac14 & \frac14 & \frac14 & \frac14 \\ -\frac18 & \frac18 & -\frac38 & \frac38 & -\frac18 & \frac18 \\ -\frac{3}{4\sqrt{15}} & -\frac{3}{4\sqrt{15}} & \frac{5}{4\sqrt{15}} & \frac{5}{4\sqrt{15}} & \frac{-3}{4\sqrt{15}} & \frac{-3}{4\sqrt{15}} \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ 6 \\ -6 \\ 0 \\ -3 \end{pmatrix}$$

$$\begin{pmatrix} \frac14 & -\frac38 & \frac{5}{4\sqrt{15}} \\ 0 & \frac14 & -\frac{3}{\sqrt{15}} \\ 0 & 0 & \frac{1}{\sqrt{15}} \end{pmatrix}$$

4 4 3 3 1 1

$$= \begin{pmatrix} -\frac{3}{64} & \frac{3}{64} & \frac{5}{192} & \frac{59}{192} & -\frac{3}{64} & \frac{3}{64} \\ -\frac{29}{160} & \frac{-19}{160} & \frac{5}{32} & \frac{11}{32} & \frac{-29}{160} & \frac{-19}{160} \\ -\frac{1}{20} & -\frac{1}{20} & \frac{1}{12} & \frac{1}{12} & -\frac{1}{20} & -\frac{1}{20} \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ 6 \\ -6 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} -\frac{63}{32} \\ -\frac{57}{80} \\ 0.2 \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$\therefore \text{Curve} \quad y = -\frac{63}{32} - \frac{57}{80}x + 0.2x^2$$

if this comes up in the exam I am throwing myself off the tower of Durham cathedral

# Spectral Decomposition

An orthogonal matrix is an $n \times n$ matrix whose columns or rows form an orthonormal basis in $\mathbb{R}^n$.

For an $n \times n$ orthogonal matrix $Q$:

$$Q^T = Q^{-1} \iff Q^T Q = I$$

$Q^T = Q^{-1}$ is orthogonal

$\det(Q) = 1$

$PQ$ is orthogonal for another $n \times n$ orthogonal matrix $P$.

$$\left.\begin{array}{l} \|Q\underline{x}\| = \|\underline{x}\| \\ \langle Q\underline{x}, Q\underline{y} \rangle = \langle \underline{x}, \underline{y} \rangle \end{array}\right\} \underline{x}, \underline{y} \in \mathbb{R}^n$$

Orthogonal Diagonalisation:

An $n \times n$ matrix $A$ is diagonalisable iff an invertible matrix $P$ exists such that

$$P^{-1}AP = D$$

Where $D$ is a diagonal matrix.

This is true if and only if $A$ has $n$ linearly independent eigenvectors.

If $P$ can be chosen to be an orthogonal matrix, then $A$ is orthogonally diagonalisable, so:

$$P^T A P = D \qquad (\text{Since } P^T = P^{-1} \text{ for orthogonal } P)$$

Spectral Theorem: For an $n \times n$ matrix $A$, the following are equivalent:

- $A$ is orthogonally diagonalisable
- $A$ has an orthonormal set of $n$ eigenvectors
- $A$ is symmetric.

Note: All symmetric matrices have all real eigenvalues, and eigenvectors from different eigenspaces are always orthogonal.

Orthogonal diagonalisation process for an orthogonally diagonalisable matrix $A$:

① (as with standard diagonalisation) Find the eigenvalues and a basis for each eigenspace of $A$

② Since eigenvectors from different eigenspaces are always orthogonal for orthogonally diagonalisable matrices, we only need to orthogonalise each basis using Gram-Schmidt and normalisation to produce an orthonormal set of eigenvectors for $A$.

③ Form the matrix $P$, whose columns are the orthonormal eigenvectors obtained in ②, and use this to find $D = P^T A P$

Example

Orthogonally diagonalise $A = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{pmatrix}$

① Find eigenvalues

$$\det(A - \lambda I) = 0$$

$$\det\begin{pmatrix} 4-\lambda & 2 & 2 \\ 2 & 4-\lambda & 2 \\ 2 & 2 & 4-\lambda \end{pmatrix} = 0$$

$$(4-\lambda)\left(\left((4-\lambda)^2-4\right)-2\left((4-\lambda)\times 2-4\right)+2\left(4-2(4-\lambda)\right)\right)=0$$
$$(4-\lambda)^3-4(4-\lambda)-4(4-\lambda)+8+8-4(4-\lambda)=0$$
$$-\lambda^3+12\lambda^2-48\lambda+64-16+4\lambda-16+4\lambda+16-16+4\lambda=0$$
$$-\lambda^3+12\lambda^2-36\lambda+32=0$$
$$\lambda^3-12\lambda^2+36\lambda-32=0$$
$$(\lambda-2)(\lambda^2-10\lambda+16)=0$$
$$(\lambda-2)(\lambda-2)(\lambda-8)=0$$

Find bases of eigenspaces

$\lambda=2$

$$(A-\lambda I)\underline{v}=\underline{0} \qquad \underline{v}=\begin{pmatrix}x\\y\\z\end{pmatrix}$$

$$\begin{pmatrix}2&2&2\\2&2&2\\2&2&2\end{pmatrix}\underline{v}=\underline{0}$$

Gauss-Jordan

$$\left(\begin{array}{ccc|c}1&1&1&0\\0&0&0&0\\0&0&0&0\end{array}\right)$$

$$x+y+z=0$$

let $y=\mu\in\mathbb{R}$
$z=\varphi\in\mathbb{R}$
$x=-\mu-\varphi$

$$\text{Eigenspace}=\text{Span}\left(\mu\begin{pmatrix}-1\\1\\0\end{pmatrix}+\varphi\begin{pmatrix}-1\\0\\1\end{pmatrix}\right)$$

$\therefore$ basis $\begin{pmatrix}-1\\1\\0\end{pmatrix},\begin{pmatrix}-1\\0\\1\end{pmatrix}$

Gram-Schmidt: $\underline{u}_1=\begin{pmatrix}-1\\0\\1\end{pmatrix},\ \underline{u}_2=\begin{pmatrix}-1\\0\\1\end{pmatrix}$

$$\underline{v}_1=\begin{pmatrix}-1\\1\\0\end{pmatrix}$$

$$\underline{v}_2=\begin{pmatrix}-1\\0\\1\end{pmatrix}-\frac{\langle\underline{u}_2,\underline{v}_1\rangle}{\|\underline{v}_1\|^2}\underline{v}_1$$
$$=\begin{pmatrix}-1\\0\\1\end{pmatrix}-\frac{1}{2}\begin{pmatrix}-1\\1\\0\end{pmatrix}=\begin{pmatrix}-1/2\\-1/2\\1\end{pmatrix}$$

Normalizing gives orthonormal basis $\begin{pmatrix}-1/\sqrt{2}\\1/\sqrt{2}\\0\end{pmatrix},\begin{pmatrix}-1/\sqrt{6}\\-1/\sqrt{6}\\2/\sqrt{6}\end{pmatrix}$

$\lambda=8$

$$(A-\lambda I)\underline{v}=0$$

$$\begin{pmatrix}-4&2&2\\2&-4&2\\2&2&-4\end{pmatrix}\underline{v}=\underline{0}$$

Gauss-Jordan

$$\left(\begin{array}{ccc|c}1&-1/2&-1/2&0\\1&-2&1&0\\1&1&-2&0\end{array}\right)\to\left(\begin{array}{ccc|c}1&-1/2&-1/2&0\\0&-3/2&3/2&0\\0&3/2&-3/2&0\end{array}\right)\to\left(\begin{array}{ccc|c}1&-1/2&-1/2&0\\0&1&-1&0\\0&0&0&0\end{array}\right)$$
$$\to\left(\begin{array}{ccc|c}1&0&-1&0\\0&1&-1&0\\0&0&0&0\end{array}\right)$$

$x-z=0$
$y-z=0$
let $z=\mu\in\mathbb{R}$
$x=y=z$

$\therefore$ basis of eigenspace: $\begin{pmatrix}1\\1\\1\end{pmatrix}$

Only one vector so no need for Gram-Schmidt, and we know it is orthogonal to the basis of all other eigenspaces of $A$ since $A$ is symmetric.

Normalized: $\begin{pmatrix}1/\sqrt{3}\\1/\sqrt{3}\\1/\sqrt{3}\end{pmatrix}$

$$\therefore \text{Matrix } P=\begin{pmatrix}-1/\sqrt{2}&-1/\sqrt{6}&1/\sqrt{3}\\1/\sqrt{2}&-1/\sqrt{6}&1/\sqrt{3}\\0&2/\sqrt{6}&1/\sqrt{3}\end{pmatrix}$$

$$D=P^TAP=\begin{pmatrix}-1/\sqrt{2}&1/\sqrt{2}&0\\-1/\sqrt{6}&-1/\sqrt{6}&2/\sqrt{6}\\1/\sqrt{3}&1/\sqrt{3}&1/\sqrt{3}\end{pmatrix}\begin{pmatrix}4&2&2\\2&4&2\\2&2&4\end{pmatrix}\begin{pmatrix}-1/\sqrt{2}&-1/\sqrt{6}&1/\sqrt{3}\\1/\sqrt{2}&-1/\sqrt{6}&1/\sqrt{3}\\0&2/\sqrt{6}&1/\sqrt{3}\end{pmatrix}$$
$$=\begin{pmatrix}2&0&0\\0&2&0\\0&0&8\end{pmatrix}$$

Spectral decomposition:

Since $D = P^T A P$, with $P^T = P^{-1}$, we can rewrite $A$ as

$$A = PDP^T$$

This is the spectral decomposition of $A$.